DATA ANALYSIS SCRIPT

Direct Classroom & Lab Reading Script (10 Problems)

This document is a direct, casual, and clear presentation script for the 10 data analysis tasks. It removes formal greetings and fluff (like 'ladies and gentlemen') and uses a direct 'here we have the data of xxx, we are doing this because of xxx' style. You can read it directly from the paper or screen to your teacher/professor during your presentation.

Each slide contains:

- Slide Title & Objective
- Key numbers and stats to point out on screen
- The exact spoken script to read out loud
- Presenter tips to guide your flow

PROBLEM 1: Problem 1: Die Rolls and Empirical Distribution

WHY WE ARE DOING THIS & OBJECTIVE

Compare 1000 simulated die rolls with the theoretical uniform distribution of a fair die.

NUMBERS / STATS TO POINT OUT ON SLIDE

1000 rolls. Counts: 1->157, 2->174, 3->151, 4->184, 5->158, 6->176.
P(even) = 0.534, P(>=5) = 0.334, P(=6) = 0.176. Chi-squared stat = 5.128 (critical value = 11.07).

EXACT READING SCRIPT

Here we have the data of 1000 simulated die rolls. We are doing this because we want to see how the empirical distribution of our rolls compares to the theoretical distribution of a fair die. First, one row in this dataset represents a single roll, showing the trial number and the face value. We counted the absolute frequencies: 1 came up 157 times, 2 came up 174, and so on. This gives us empirical probabilities: the probability of getting an even number is 0.534, at least a 5 is 0.334, and exactly a 6 is 0.176. A fair die should have a flat probability of 1/6, which is about 16.7% for each face. The fluctuations we see in our frequencies are just natural random variations that occur in any finite sample. We ran a Chi-Squared goodness-of-fit test and got a statistic of 5.128. Since this is below the critical threshold of 11.07, mathematically we cannot reject the hypothesis that the die is fair. However, in reality, we know the die was generated with slightly biased probabilities, which shows that a sample size of 1000 is not always large enough to detect subtle biases.

PRESENTER ACTION / TIP: Point to the bar chart on your slide and show the fluctuations around the 16.7% line.

PROBLEM 2: Problem 2: Coin Tosses and Relative Frequencies

WHY WE ARE DOING THIS & OBJECTIVE

Track relative frequencies over 2000 coin tosses to demonstrate the Law of Large Numbers.

NUMBERS / STATS TO POINT OUT ON SLIDE

2000 tosses. Heads = 1038 (51.9%), Tails = 962 (48.1%).
True probability used in generator: Heads = 0.52, Tails = 0.48.

EXACT READING SCRIPT

Here we have the data of 2000 coin tosses. We are doing this to show the Law of Large Numbers in action. One row is just one toss showing whether it landed heads or tails. Overall, we got 1038 heads, which is 51.9%, and 962 tails, which is 48.1%. If you look at the plot of relative frequency against trials, you can see that the rate of heads fluctuates heavily at the beginning when the sample size is small. But as we get more trials, the curve stabilizes and gets very close to the true probability, which is 52% in this simulation. This proves that empirical relative frequency converges to the true theoretical probability as the number of trials increases, and it shows that the coin is slightly biased towards heads.

PRESENTER ACTION / TIP: Trace the s-curve or stabilization line on the plot with your mouse/pointer to show the sönümlenme (dampening of noise).

PROBLEM 3: Problem 3: Exam Scores in Two Groups

WHY WE ARE DOING THIS & OBJECTIVE

Compare performance of student groups A and B using summary statistics.

NUMBERS / STATS TO POINT OUT ON SLIDE

Overall mean = 68.22, median = 69.00.
Group A: mean = 72.15, median = 73.00, std dev = 11.45, pass rate = 85%.
Group B: mean = 64.29, median = 65.00, std dev = 15.12, pass rate = 71%.

EXACT READING SCRIPT

Here we have the exam scores of two student groups, A and B. We are doing this because we want to see how their performances compare and why looking only at averages can be misleading. Group A has a mean of 72.15 and Group B has a mean of 64.29. Group A looks better at first glance. But if you look at the standard deviation, Group A is 11.45 while Group B is 15.12. This means Group B's scores are much more spread out, having both very high scores and very low scores, while Group A is much more consistent. Because B has a lot of low scores and high variance, their pass rate is only 71% compared to Group A's 85%. This shows that averages alone don't tell the whole story; you need to analyze the variance to understand the spread.

PRESENTER ACTION / TIP: Point at the boxplots of both groups. Show how Group B's box is wider, indicating higher variability.

PROBLEM 4: Problem 4: Delivery Times, Skewness, and Outliers

WHY WE ARE DOING THIS & OBJECTIVE

Analyze skewed delivery times and identify outliers using the 1.5*IQR rule.

NUMBERS / STATS TO POINT OUT ON SLIDE

120 deliveries. Mean = 38.45 min, Median = 32.10 min.
Q1 = 25.50 min, Q3 = 45.80 min, IQR = 20.30 min.
Outlier upper bound = 76.25 min. Outliers found = 8. Gecikme rate (>50 min) = 17.5%.

EXACT READING SCRIPT

Here we have the delivery times of 120 orders. We are doing this because we want to analyze the skewness of our delivery process and identify outlier deliveries. The mean delivery time is 38.45 minutes, but the median is 32.10 minutes. The mean is higher because the distribution is right-skewed, meaning we have a few very long deliveries that pull the average up. We calculated Q1 and Q3, and using the 1.5 IQR rule, we found that any delivery taking over 76.25 minutes is an outlier. We found 8 outliers in this dataset. We also have a delay rate of 17.5% for deliveries over 50 minutes. This analysis shows why the median is a much more robust and informative metric than the mean when you are dealing with skewed data like delivery times.

> **PRESENTER ACTION / TIP: Point to the long right tail on the histogram to explain skewness, and highlight the 8 outlier points on the boxplot.**

PROBLEM 5: Problem 5: Customer Survey and Conditional Frequencies

WHY WE ARE DOING THIS & OBJECTIVE

Analyze customer satisfaction and renewals using conditional probabilities.

NUMBERS / STATS TO POINT OUT ON SLIDE

500 customers. Overall renewal rate = 64.2%.
Channels: Email (72%), Phone (58%), Web (66%).
P(renewed | sat=5) = 88.5%, P(sat=5 | renewed) = 35.6%.

EXACT READING SCRIPT

Here we have customer satisfaction and renewal data from 500 accounts. We are doing this to analyze the relationship between customer satisfaction scores and renewal rates using conditional probabilities. The overall renewal rate is 64.2%. If we look at the channels, Email has the highest renewal rate at 72% and Phone has the lowest at 58%. The conditional probability P(renewed given satisfaction equals 5) is 88.5%, which shows that highly satisfied customers are very likely to renew. But P(satisfaction equals 5 given renewed) is only 35.6%. This is because many customers who renewed actually had lower satisfaction scores, like 3 or 4. These two conditional probabilities answer completely different questions and shouldn't be confused.

PRESENTER ACTION / TIP: Emphasize that the order of conditional probability matters: one predicts renewal from satisfaction, the other details the satisfaction of those who renewed.

PROBLEM 6: Problem 6: Factory Measurements and Specification Limits

WHY WE ARE DOING THIS & OBJECTIVE

Evaluate part measurements from two machines against tolerance limits (98mm - 102mm).

NUMBERS / STATS TO POINT OUT ON SLIDE

Target = 100 mm. Spec limits = 98.00 to 102.00 mm. Overall within-spec rate = 92.4%.
Machine 1: mean = 100.02 mm, std dev = 0.45 mm, within-spec = 99.2%.
Machine 2: mean = 99.85 mm, std dev = 1.20 mm, within-spec = 82.5%.

EXACT READING SCRIPT

Here we have the length measurements of parts produced by two machines. We are doing this because we want to see which machine is better at keeping parts within the spec limits of 98 to 102 mm. Overall, 92.4% of all parts are within spec. But if we split the data by machine, Machine 1 has a mean of 100.02 mm and a standard deviation of 0.45 mm, which puts 99.2% of its parts within spec. Machine 2 has a mean of 99.85 mm—which is close to target—but its standard deviation is 1.20 mm. Because of this high variability, only 82.5% of its parts are within spec, meaning 17.5% is waste. This proves that having a good average is not enough if your process variance is too high.

PRESENTER ACTION / TIP: Point out that Machine 2 needs recalibration or maintenance because its standard deviation (1.20 mm) is too high.

PROBLEM 7: Problem 7: Call Center Requests Over Time

WHY WE ARE DOING THIS & OBJECTIVE

Model hourly call volumes and assess suitability of a Poisson distribution.

NUMBERS / STATS TO POINT OUT ON SLIDE

Hourly mean = 14.2, variance = 15.1.
Weekday mean = 16.5, Weekend mean = 8.4.
Peak call volume hours: 10:00 - 14:00.

EXACT READING SCRIPT

Here we have hourly call volumes at a call center over 30 days. We are doing this because we want to check if call arrivals follow a Poisson distribution. The overall mean call volume is 14.2 per hour, and the variance is 15.1. Since the mean and variance are very close, a Poisson model is a good fit. But we shouldn't treat the whole dataset as one single Poisson process because the arrival rate is not constant. Weekdays have a mean of 16.5 calls per hour, while weekends have only 8.4. Also, call volume peaks heavily between 10 AM and 2 PM. So we should model this as a non-homogeneous Poisson process to schedule staff correctly for peak hours.

PRESENTER ACTION / TIP: Mention that staff scheduling based on the overall mean (14.2) would fail during peak hours when volume is much higher.

PROBLEM 8: Problem 8: Waiting Times and Empirical CDF

WHY WE ARE DOING THIS & OBJECTIVE

Compare standard vs priority wait times and use ECDF to estimate probabilities.

NUMBERS / STATS TO POINT OUT ON SLIDE

Overall mean = 11.2 min, median = 8.5 min.
Priority: mean = 4.1 min, median = 3.2 min.
Standard: mean = 13.5 min, median = 11.2 min.
ECDF estimations: $P(T <= 5) = 45\%$, $P(T <= 10) = 68\%$, $P(T > 20) = 12\%$.

EXACT READING SCRIPT

Here we have the waiting times for standard and priority services. We are doing this because we want to build an Empirical CDF to estimate wait probabilities. Priority wait time has a median of 3.2 minutes, while standard has 11.2 minutes. The ECDF plot lets us read probabilities directly from the data without assuming a distribution. For example, the probability of waiting 5 minutes or less is 45%, 10 minutes or less is 68%, and the probability of waiting more than 20 minutes is only 12%. This helps us set realistic service level agreements.

> **PRESENTER ACTION / TIP: Point at the ECDF curve and show how to read the probabilities on the y-axis for a given wait time on the x-axis.**

PROBLEM 9: Problem 9: Warehouse Orders, Demand, and Returns

WHY WE ARE DOING THIS & OBJECTIVE

Analyze order quantities and return rates across product groups and warehouses.

NUMBERS / STATS TO POINT OUT ON SLIDE

Total quantity = 45,210. High demand: Electronics (18,200), Apparel (12,400).
Overall return rate = 14.8%. Apparel return rate = 22.1%. Northern Warehouse return rate = 16.2%.

EXACT READING SCRIPT

Here we have orders and return rates for 5 product groups across 3 warehouses. We are doing this to identify operational risks and demand distribution. Total quantity ordered is 45,210. Electronics has the highest demand with 18,200 units, and Apparel has 12,400. But the overall return rate is 14.8%, and Apparel has a huge return rate of 22.1%. Also, the Northern Warehouse has the highest return rate at 16.2%. This shows that high sales in apparel come with a high return cost, and Northern Warehouse processes need to be checked.

PRESENTER ACTION / TIP: Conclude that focus should be placed on reducing Apparel returns, perhaps by improving sizing guides.

PROBLEM 10: Problem 10: Correlation Traps

WHY WE ARE DOING THIS & OBJECTIVE

Demonstrate how outlier groups distort correlation coefficients and why scatter plots are necessary.

NUMBERS / STATS TO POINT OUT ON SLIDE

Full correlation (r) = 0.82 (strong positive relationship).
Without outlier group (r) = -0.15 (weak negative relationship).
Finding: The outlier group distorts the correlation math.

EXACT READING SCRIPT

Here we have X and Y variables from a dataset of 100 observations. We are doing this to show how correlation coefficients can be highly misleading. If we calculate the overall correlation coefficient, we get a strong positive value of r = 0.82. But if you look at the scatter plot, the main body of data actually has a weak negative trend of r = -0.15. The positive correlation is completely caused by a small outlier group in the top right corner. This proves that you should always look at the scatter plot instead of just trusting the correlation number.

PRESENTER ACTION / TIP: Point to the two scatter plots side by side to show how removing the outlier group completely flips the trend from positive to negative.